

Chapter: Magnets

1. Introduction to Magnets

A **magnet** is any material that has the property of attracting certain substances like iron, cobalt, and nickel. This attractive property is called **magnetism**.

The Story of Discovery (Magnes the Shepherd)

The word "magnet" originates from ancient Greece.

- A shepherd named **Magnes** was tending his sheep in a region called **Magnesia**.
- The iron casing on the tip of his wooden walking stick and the nails in his sandals suddenly stuck to a massive black rock.
- That natural rock was **Magnetite** (Fe_3O_4), an ore of iron and the world's first discovered natural magnet. It also became known as **lodestone** (meaning "leading stone") because it was used for navigation.

2. Types of Magnets

Magnets are broadly classified into two categories based on their origin and how long they keep their magnetic properties:

A. Natural Magnets

- Found directly in nature.
- They have irregular shapes and weak magnetic strength.
- **Example:** Magnetite (Lodestone).

B. Artificial Magnets

- Man-made magnets created from pieces of iron, steel, or special alloys.
- They can be shaped cleanly and made incredibly strong.

As seen in the diagram above, artificial magnets are engineered into specific shapes depending on how they will be used:

- **Bar Magnet:** A simple rectangular block.
- **Horseshoe Magnet:** Shaped like a "U" to bring both magnetic poles close together, doubling its holding strength.
- **Cylindrical/Ball-ended Magnet:** Thin rods ending in small spheres.

3. Magnetic vs. Non-Magnetic Materials

Not everything reacts to a magnet. We split all matter into two functional camps:

Property	Magnetic Materials	Non-Magnetic Materials
Definition	Materials that are strongly attracted by a magnet.	Materials that show absolutely no attraction to a magnet.
Can be Magnetized?	Yes, they can be turned into temporary or permanent artificial magnets.	No, they can never hold a magnetic charge.
Examples	Iron, Steel, Nickel, Cobalt.	Wood, Plastic, Gold, Copper, Aluminium, Paper.

4. The Poles of a Magnet

If you roll a bar magnet in a pile of iron filings, you will notice a strange thing: the filings gather heavily at the two ends, leaving the middle almost completely bare.

- The regions near the ends of a magnet where the magnetic attraction is at its maximum strength are called **Magnetic Poles**.
- Every single magnet has exactly **two poles**:
 1. **North-Seeking Pole (North Pole / N)**: The end that naturally points toward the Earth's geographic North when suspended freely.
 2. **South-Seeking Pole (South Pole / S)**: The end that points toward the Earth's geographic South.

The Law of Magnetic Poles:

- **Like poles repel**: North repels North ($N \rightarrow N$); South repels South ($S \rightarrow S$).
- **Unlike poles attract**: North strongly attracts South ($N \rightarrow S$).

Critical Concept: Magnetic Monopoles Do Not Exist

You cannot have a single, isolated pole. If you snap a bar magnet exactly in half right down the middle, you do *not* get an isolated North piece and an isolated South piece. Instead, two brand-new poles instantly form at the break line, leaving you with two smaller, fully functional magnets—each containing its own N and S pole.

5. Finding Directions: The Magnetic Compass

Because a freely suspended magnet always swings around until it aligns itself with the Earth's North-South direction, it has been used for navigation for centuries.

The **Magnetic Compass** is a simple device consisting of a small glass-topped aluminum casing housing a magnetized needle balanced perfectly on a central pivot point.

- The needle is entirely free to rotate horizontally.
- The North pointer of the needle is usually painted a distinct color (like red or blue, as shown in the visual above).
- To find directions, the traveler rotates the outer dial casing until the "N" mark on the card aligns directly underneath the colored pointer of the needle. Once aligned, all other directions (East, West, South) are immediately visible on the dial face.

6. How to Make Your Own Magnet (Single Touch Method)

You can temporarily turn a plain piece of iron into a working magnet using the **Single Touch Method**:

1.Position the Iron Bar:Step 1.

Place a plain, non-magnetized iron bar flat on a table surface.

2.Establish Contact:Step 2.

Take a strong permanent bar magnet. Place one of its poles (e.g., the North Pole) down onto one edge of the iron bar.

3.Stroke the Bar:Step 3.

Without lifting the magnet, drag it firmly along the surface of the iron bar all the way to the opposite end.

4.Lift and Loop:Step 4.

Lift the bar magnet high into the air, bring it back to the original starting edge, and place the *exact same pole* down again.

5.Repeat:Step 5.

Repeat this continuous looping stroke 30 to 40 times in one direction only. The iron bar will now attract small steel pins!

7. Demagnetization & Caring for Magnets

Magnets are not indestructible. They can easily lose their magnetic properties—a process called **demagnetization**—if handled poorly.

How Magnets Lose Magnetism:

- **Heating:** Exposing a magnet to high temperatures disrupts its internal atomic alignment.
- **Hammering or Dropping:** Repeatedly dropping a magnet onto a hard floor or hitting it violently with tools shakes its magnetic structural order loose.
- **Improper Storage:** Leaving magnets loose near each other allows conflicting poles to cancel one another out over time.

How to Store Magnets Properly:

- **Bar Magnets:** Should always be stored in pairs with their **unlike poles facing each other** (North next to South). They must be separated by a small strip of insulating wood, and their open ends joined firmly using pieces of soft iron called **magnetic keepers**.
- **Horseshoe Magnets:** Only require a single piece of soft iron keeper placed flat across both poles.

Multiple Choice Questions (Practice Test)

Test your knowledge with these targeted practice questions.

Q1. Which of the following is a non-magnetic material?

- A) Iron
- B) Cobalt
- C) Aluminium
- D) Nickel

Q2. A freely suspended bar magnet always comes to rest pointing along which direction?

- A) East - West
- B) North - South
- C) North - East
- D) South - West

Q3. What happens if a bar magnet is broken cleanly into three small pieces?

- A) One piece gets only a North pole, one gets a South pole, and one loses magnetism.
- B) All three pieces become independent, complete magnets with both N and S poles.
- C) The pieces lose their magnetic power entirely.
- D) Only the end pieces remain magnetic.

Q4. Two magnets are placed near each other. Their North poles are facing one another directly. What will they do?

- A) Attract each other strongly
- B) Rotate by 90 degrees
- C) Repel each other away
- D) Cancel each other out and turn into normal iron

Q5. To protect bar magnets from losing their strength over time, they should be stored:

- A) Wrapped securely in thick plastic bags
- B) In pairs with like poles side-by-side
- C) In pairs with unlike poles side-by-side, separated by wood, with iron keepers across the ends
- D) Submerged completely in cold water

Answer Key & Explanations

1. **(C) Aluminium** — Iron, cobalt, and nickel are the three primary magnetic elements; aluminium does not react to magnetic fields.
2. **(B) North - South** — The Earth's magnetic field naturally forces a free-floating or suspended magnet to align with its geographic North and South poles.
3. **(B) All three pieces become independent...** — Magnetic monopoles cannot exist in nature. Breaking a magnet creates brand-new pairs of poles at the break sites.
4. **(C) Repel each other away** — Like magnetic poles (\$N-N\$ or \$S-S\$) always exert a repulsive force against one another.
5. **(C)** — Proper storage with wood dividers and soft iron keepers prevents self-demagnetization by keeping the magnetic loops closed securely.